

Analyzing Wealth Inequality in 1427 Renaissance Florence through Tax Data

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Introduction

Renaissance Florence is one of the most well-documented historical economies in the world. This can be attributed to its unique history in which arts and trade guild members were exclusively elected into political offices. This was the result of a 1293 constitution called the Ordinances of Justice, which excluded nobility from political power (Brown University, 2002). The most wealth and political power was held by seven major guilds, which were called the Arti Maggiori (Becker, 2019).

In the 15th century, the city introduced an extensive tax survey on all households in the city, known as the Catasto. The Catasto was seen as a way to improve the fairness and efficiency of the tax system through its extensiveness (McLean, 2005). This produced a detailed dataset of property and business assets for each household, alongside demographic information such as the occupation and sex of the heads of the households (Herlihy & Klapisch-Zuber, 2002). Given the richness of this data, various historians have collected and digitized Catasto tax records of the years 1403, 1427, 1457 and 1480 (Belloc, Drago, Fochesato, & Galbiati, 2024b).

For this assignment, the Catasto data will be implemented in a Bayesian Network to create a model that can be used to study wealth inequality. To demonstrate the effectiveness of the resulting model, the following research question will be investigated:

Did trade guild membership result in a higher amount of taxable assets?

Overview of the data

For the 2024 study Multigenerational Transmission of Wealth: Florence 1403-1480 (Belloc et al., 2024b), the Catasto data of 1403, 1427, 1457 and 1480 (Belloc, Drago, Fochesato, & Galbiati, 2024a) has been collected. For this assignment, the 1427 data will be used because this year contains interesting variables that are not present in the other datasets, such as marital status and housing arrangement.

The 1427 Catasto contains the tax collections of more than 9,000 Florentine households, and includes the variable *trade* that can be assigned 99 possible professions. To simplify the model, various variables such as names and location (in the city of Florence) will be omitted. The *trade* variable will be simplified to denote whether someone was a

member of the Arti Maggiori or not. Nine variables have been chosen for the Bayesian Network, which can be found in table 1.

Variable	Type	Domain/Categories	Description
Age	Ordinal	{child [0, 18), young_adult [18, 30), adult [30, 50), senior [50, 70), elder [70, ∞)}	Age
Sex	Categorical	{male, female}	Gender
Marital_status	Categorical	{single, engaged, married, separated, widowed}	Matrimonial status
Housing_arr	Categorical	{owner, tenant, with_family}	Housing arrangement
Household_size	Ordinal	{1-2, 3-4, 5-7, 8-12, 13+}	Number of people in the household
Trade_guild	Categorical	{yes, no}	Arti Maggiori membership
Investments	Ordinal	{poor [0, 29), lower middle [29, 210), upper middle [210, 893), wealthy [893, 2634), affluent [2634, 14299), elite [14299, ∞)}	Real estate, private investments and public investments in Florins
Deductions	Ordinal	{poor [0, 30), lower middle [30, 90), upper middle [90, 224), wealthy [224, 597), affluent [597, 3408), elite [3408, ∞)}	Tax deductions in Florins (debts and charges)
Taxable	Ordinal	{poor [0, 1), lower middle [1, 116), upper middle [116, 638), wealthy [638, 1931), affluent [1931, 11208), elite [11208, ∞)}	Taxable assets after deductions in Florins

Table 1: *The variables in the Bayesian Network with their domains after discretization.*

Variables **investments**, **deductions** and **taxable** were discretized with percentile cutoffs binned at 0.25, 0.50, 0.75, 0.90 and 0.99, measured in Florins.

Sex has been chosen as a variable because women played a different role in Renaissance society than men. They were typically expected to take care of the household and let their husband be the provider (Romano, 1989). They were generally not granted full guild membership (Hurlburt, 2015). However, wealthy married women have also been found to frequently own property (Smith, 1998). Women were excluded from owning immobile property, but were usually given a portion of the family inheritance as a dowry when they left the family to marry (Chabot, 2006).

Household_size has been chosen as a variable because upon study of the 1426 Catasto, wealthy families have been found to be considerably larger than those of the poor (Grubb, 2019). **Housing_arr** was chosen because house ownership was seen as a symbol of wealth in the community, and is a directly measurable asset (Chabot, 2006). Taxable assets indicate the wealth of a household, and this field is dictated by investments and deductions.

Causal dependencies

The data has been implemented in a Bayesian Network to capture the relationships between influencing factors for household wealth, which can ultimately be expressed as taxable assets. This resulted in a causal model, with the following causalities:

In the network, sex was put before marital_status because the conditional probability of marriage given sex varies at least by age (Grubb, 2019) and social status (Padgett, 2010). Since women were rarely granted full guild membership, trade_guild depends on sex as well (Hurlburt, 2015). Aside from Age affecting marriage behavior, it also dictated whether someone was old enough to participate in a trade and housing situation. This is significant because some children are also included in the dataset, for instance if they were family heirs.

Guild membership may provide access to more profitable business opportunities, and therefore to more investments. If they invested in real estate, their housing arrangement may change. Having more investments also means being able to support more children, resulting in an increase in household size. Household size and housing arrangement result in deductions, as both variables result in costs.

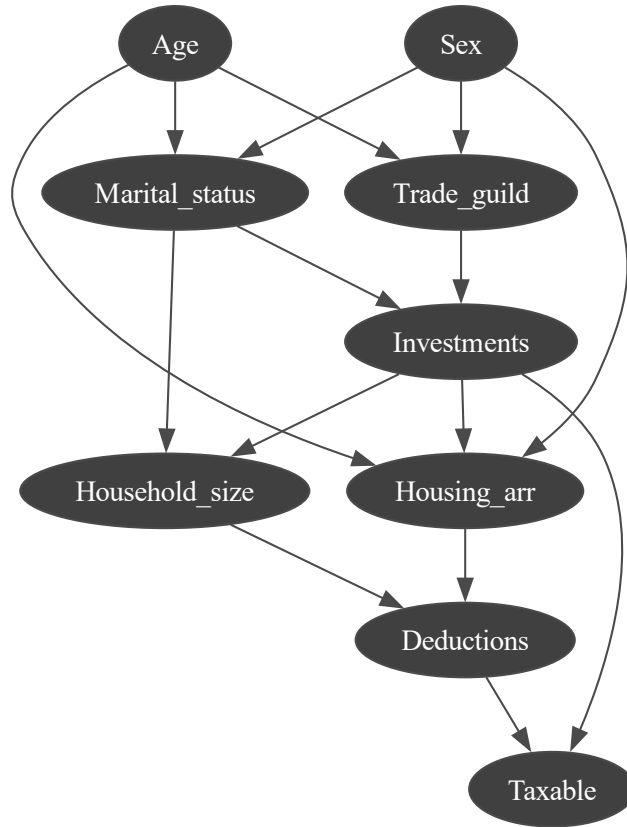


Figure 1. Bayesian Network of the causal dependencies.

Prior probabilities

To apply data-driven probabilities directly derived from the dataset onto the Bayesian Network, `learnParameters()` is used. The prior probabilities are subsequently found using the `showInference()` command in pyAgrum, and can be viewed in figure 2. An roughly equal distribution can be seen for **Investments** and **Deductions**, which indicates that the distribution by percentiles was successful. In **Taxable**, it can be seen that a large part of the population had 0 taxable assets, meaning they had either no investments or their deductions were equal to or higher than their investments.

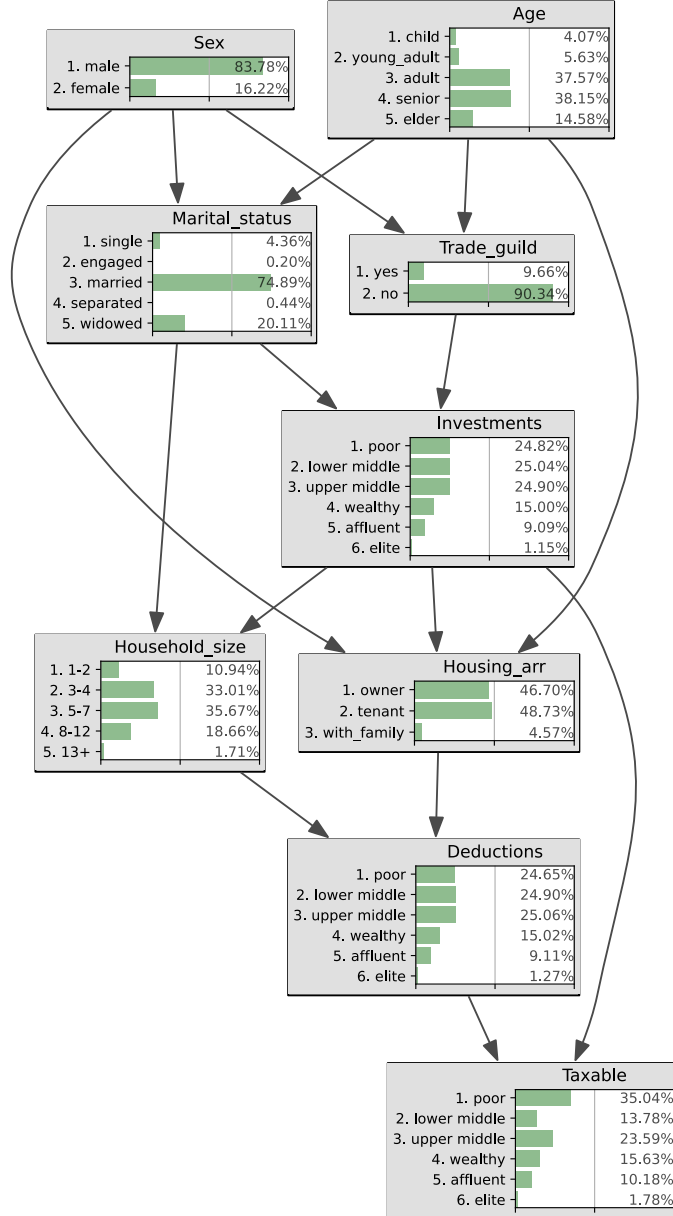


Figure 2. Prior probabilities using pyAgrum inference on the dataset.

Inference

As can be seen under `trade_guild`, less than 10% of household heads were members of the Arti Maggiori. To find out what the wealth distribution exclusively for members of the Arti Maggiori looks like, the Lazy Propagation inference method is used with the variable `trade_guild = yes` (sample size 685). Lazy Propagation avoids unnecessary computations by only using relevant parts of the network (Madsen & Jensen, 2013).



Figure 3. Posterior distribution of taxable assets for members of the Arti Maggiori.

The result can be viewed in figure 3. Arti Maggiori membership is shown to be economically advantageous, which matches the historical accounts (Becker, 2019). There are less people with no taxable assets than before, Notably, even among the Arti Maggiori there is still a small elite holding vast amounts of wealth.

Finally, as illustrated in figure 4, a more complex scenario was studied: the wealth distribution of widowed women who owned a house (sample size 508).



Figure 4. Posterior distribution of taxable assets for widowed female house-owners.

Discussion

This model illustrates how Bayesian Networks can be used to study historical economical dynamics. A limitation of this method is that this model only provides a snapshot of a single year, 1427. By also constructing models over other years, more detailed insights in socio-economic developments can be derived.

Another limitation is the necessary discretization of continuous variables, such as investments, household size and age, which may make more nuanced assertions difficult to study. Furthermore, the causal flow of the variables is also necessarily simplified. For instance, wealth may play a role in marital status as well, since wealthier families may have had different marriage strategies than poorer families. Likewise, being more wealthy may have made guild membership more accessible. Such feedback effects cannot be implemented using Bayesian Networks, as they do not allow causal loops.

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